

# **Alzheimer's Disease Detection Using Multimodal Modality**



**By:**

**Durab Haider**

**46614**

**Muhammad Umar**

**46952**

**Supervised by:**

**Mr. Mubariz Rehman**

**Faculty of Computing**

**Riphah International University, Islamabad**

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## Final Approval

This is to confirm that the report prepared by ***Durab Haider (46614)*** and ***Muhammad Umar (46952)*** has been evaluated to fulfill the requirement for the Bachelor of Science in Computer Science (BSCS). It is hereby recommended that the report fulfills all criteria set by Riphah International University, Islamabad to award Bachelor of Science in Computer Science (BSCS).

### Committee:

1

---

Mr. Mubariz Rehman  
(Supervisor)

2

---

Mr. Musharraf Ahmed  
(Head of Department)



## Declaration

We hereby certify that the document entitled “**Detection of Alzheimer’s disease using Multimodal Modality**” is an original one, and there has not been any copying from any source whatsoever. We hereby further certify that the project along with the report has been developed entirely by ourselves, with the help of our instructors, especially Supervisor **Mr. Mubariz Rehman**. In case any part of the system is found to be borrowed from some other source, we will face the resultant consequences.

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**Durab Haider**

**46614**

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**Muhammad Umar**

**46952**

## **Dedication**

This assignment is hereby dedicated to our parents in honor of their limitless love and support. We are grateful to our teacher, Mr. Mubariz Rehman, for his valuable advice and motivation in this whole process. Our heartfelt thanks are due to our friends for motivating us to continue and complete this task successfully. This whole assignment could never be completed without the help of everyone involved.

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**Durab Haider**

**46614**

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**Muhammad Umar**

**46952**

# **Abstract**

Alzheimer's disease is an important neurological disease causing memory loss and other cognitive problems worldwide. Traditional methods of diagnosing Alzheimer's disease are often slow and highly dependent on the expertise of clinicians, resulting in a delay in diagnosing the disease at an early stage. This project uses artificial intelligence in order to diagnose quickly and efficiently. This proposed model uses MRI brain scans along with cognitive tests in order to classify the patients into different stages of cognitive impairment and Alzheimer's disease. The different stages include Cognitively Normal, Early Mild Cognitive Impairment, Late Mild Cognitive Impairment, and Alzheimer's disease. The aim of this project is to avoid any errors by humans, make the process timely and help in early intervention.



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## **Abstract**

The proposed system for detecting Alzheimer's disease is a smart and efficient system that enables the identification and classification of the occurrence of Alzheimer's disease at its earliest stage. The system is based on advanced deep learning techniques to process brain MRI images together with cognitive assessments for diagnosing the disease accurately. The system will be able to classify images through image processing and data analysis processes supported by artificial intelligence. There are different stages of patients which include: Cognitively Normal, Mild Cognitive Impairment and Alzheimer's disease. This system automates the whole process, thus making it fast, efficient and reducing errors made by human beings.

The system is easy to use, thus supporting interactions between doctors and researchers. It provides systematic outcomes for the users. The AI-driven framework provides for high accuracy and scalability, appropriate for real world health care use settings. This project will help to identify disease at an early stage and improve disease stage, allowing the crop to be better managed. Facilitate prompt medical interventions, enhance the care of patients and participate in contributions to the creation of smart and eco-friendly healthcare solution.

## **Chapter 1: Introduction**

## **Chapter 1: Introduction**

The Alzheimer disease (AD) is a progressive and neurodegenerative condition which is currently regarded as the main cause of global dementia. As such, the disease constitutes a major component associated with cognitive impairments linked to the aging process. Specifically, AD implies gradual deterioration of memory, executive functions and behavioral control which lead to loss of individual's autonomy and increased mortality risks. With the accelerated aging process witnessed globally, the incidence of the disease is projected to increase dramatically, thus creating additional burden on various health care systems and caregivers/relatives. So far, a wide array of studies have not brought about any effective treatments for AD; hence, early and precise diagnosis is one of the key aspects involved in the treatment of Alzheimer disease.

The conventional approach to AD diagnosis mainly includes a range of clinical methods and cognitive screening tools, as well as the visual analysis of MRI data. While there are common practices of using MoCA scale and CDR in diagnosing AD, these approaches are subject to various kinds of bias associated with the patient's level of education, cultural background and examiner's personal views. In turn, the manual analysis of MRI scans is time-consuming and prone to errors committed by observers due to subjective judgment that affects the accuracy of finding neurodegenerative markers associated with AD and MCI.

In recent times, a number of AI innovations, especially in deep learning have shown promising results in overcoming the aforementioned limitations of diagnosis of diseases. For example, CNNs have been proved to be useful in feature extraction of high dimensional data, such as structural MRIs and have found success in detection of patterns of disease. However, one limitation that has plagued many AI techniques is the reliance of data from a single modality, in most cases MRI. This could be because of the fact that Alzheimer's disease is multivariate in nature; where both the structural brain changes and cognitive changes interact in complicated manners; hence, an univariate analysis of a multivariate problem might prove to be insufficient.

Multimodal learning theories have come to fore front as a solution of this problem where multiple data sources, including neuroimaging and cognitive data have been



fused to form a predictive model. Some studies have been carried out previously which have shown that a combination of structural MRI features and clinical/neuropsychological features has resulted in much better classification of people in different states of Alzheimer's, including cognitively normal, early MCI, late MCI, and Alzheimer's disease. However, there are many difficulties associated with multimodal fusion such as data heterogeneity, alignment issues, class imbalance and explainability of models.

This project aims to fill such gaps by suggesting a multimodal deep learning model to detect early signs of the Alzheimer disease with the integration of the MRI-based features and cognitive tests scores into a single attention-based model. Deep CNNs are used to process structural MRI volumes to obtain high levels of spatial representations, whereas cognitive measures of MoCA, ADAS-13, and CDR-SB are used as additional clinical measures. The attention mechanisms are used to enhance feature learning by highlighting the diagnostically relevant areas and patterns to enhance the performance of classification and generalization.

The primary advantage of this project is the integration of the whole system design, which goes further than just developing models and covers a real-world user interface (UI). The implemented UI allows uploading MRI images and cognitive assessments for real-time predictions and visualizations of diagnostic pathways. System integration is critical for the application of results obtained at the academic stage to clinical practice. At the same time, many studies do not cover this aspect. It is expected that the developed solution will bring more usability and applicability due to the combination of cutting-edge usability with backend intelligence.

In addition to that, the appraisal methodology used in the current project relies on the application of practical neuroimaging data available in modern databases such as Alzheimer's Disease Neuroimaging Initiative (ADNI). In order to validate the clinical significance and reliability of the proposed solution, it is tested in relation to four diagnostic groups – AD, early MCI, late MCI, and cognitively normal people. The study acknowledges the limitations caused by heterogeneity and demographical bias of datasets but emphasizes the importance of reproducibility, scalability, and methodology transparency.

To conclude, the study is relevant to the sphere of medicine using AI technologies in terms of providing a strong and explainable deep learning model for diagnosing Alzheimer's disease. The combination of different sources of data (MRI and cognitive), attention-based architecture, and UI makes this solution capable of influencing diagnostic methods in the future and becoming the basis of a new clinical decision-support system.

## **1.1 Goals and Objectives**

The following goals are aimed at with this project:

1. Developing and establishing a multimodal Deep Learning program, combining structural MRI data and cognitive assessment data to detect Alzheimer disease.
2. To test and evaluate attention boosted CNN architectures having ability to identify discriminative neuroimaging markers that can be used in disease progression.
3. To classify the subjects into clinically meaningful groups, e.g., AD, LMCI, EMCI, and CN.
4. This research seeks to create an interface that would help achieve friendly communication between users and the diagnostic system while at the same time providing usability and real-time prediction in the system.
5. To evaluate the performance of the system critically using pre-established databases.

## **1.2 Scope of the Project**

The current research focuses on the development of a multimodal deep learning technique for the identification and classification of Alzheimer's disease using magnetic resonance imaging (MRI) and cognitive test information. The solution is provided with the help of the integration of medical imaging, artificial intelligence, clinical data, and clinical trials.

This process implies pre-processing and analyzing brain MRI images obtained from the privately held databases in order to detect certain patterns in the structures of brains that suffer from Alzheimer's disease and the prodromal forms of it. Deep Convolutional Neural Networks are used in order to learn imaging features in a fully automated way.

The attention model is applied to increase the significance of the features and highlight the brain areas important for the diagnosis.

The scope of the study also covers the use of traditional cognitive assessments, which are MoCA, ADAS-13, and CDR-SB. They can be used as complementary markers of cognitive decline. The cognitive variables are incorporated into MRI-based features in a multimodal learning approach for better performance in clinically meaningful subsets, which include cognitively normal (CN), early mild cognitive impairment (EMCI), late mild cognitive impairment (LMCI), and Alzheimer's disease (AD) cases.

Further, deep learning models will be utilized for model comparison based on MRI-based model and multimodal fusion models. The comparison of models will be carried out using different metrics such as accuracy, precision, recall, and F1 score. The training, validation, and testing of models will be done in a controlled experiment setting.

Another part of the project that is under consideration is the development of a user interface to allow users to input MRI images and cognitive scores and get automatic predictions on diagnosis. Thus, the scope of the project includes developing the connection between the algorithm and usability, which will inform future potential implementation of the algorithm for the research purposes or in practice.

The current project does not cover such aspects as treatment recommendations, disease prognosis, or longitudinal progression modeling. Clinical real-time verification and implementation in hospitals are not included in the scope of the research at the moment.

## **Chapter 2: Literature Review**

## **Chapter 2: Literature Review**

Here we will discuss about the literature review.

### **2.1 Introduction**

Alzheimer's disease (AD) is a neurodegenerative condition that poses an urgent public health problem in the contemporary world. The pathology evolves in a stealthy manner and in most cases, it begins several years before the manifestation of clinical symptoms thus making the early diagnosis a daunting task. The traditional diagnostic paradigms have been based to a large extent on cognitive screening tools and the subjective interpretation of neuroimaging by clinicians. These biases based on subjective measurements often cause a failure to recognize even the smallest pathological changes at the earliest stages of the disease, which means that in the end it becomes the significant percentage of the patients who will only be diagnosed at the moment when it is already too late, and the neural damage cannot be removed. The developments made in artificial intelligence, especially the deep learning (DL) branch, have revolutionized the analysis process of medical images and decision-making processes. DL models have the ability to automatically derive patterns of discrimination in high-dimensional neuroimaging scans and thus perform better than conventional machine-learning methods that require hand-defined features [1]. Furthermore, recent studies have also been more specifically insightful to multimodal DL systems, in which magnetic resonance imaging (MRI) data are utilized together with cognitive and clinical evaluations to represent the characteristics of the disease complementarily. The chapter is a critical review of the modern literature on DL-based AD detection, specifically on multimodal frameworks, thus outlining their accomplishments, weaknesses, and outstanding gaps.

### **2.2 Background and Problem Elaboration**

The pathological characteristics of AD cover several dimensions, including structural brain shrinkage, functional loss and cognitive impairment. Structural MRI is the most commonly used modality in the visualization of anatomical changes as hippocampal or cortical thinning, although it may be extraordinary fine at the early disease stage, and can have a significant inter-individual variability. Cognitive batteries (Montreal Cognitive Assessment, The Alzheimer Disease Assessment Scale – Cognitive subscale,

as well as the Clinical Dementia Rating – Sum of Boxes, offer independent information about functional decline; however, the assessment process can be affected by initial levels of education, personal biases, and patient cooperation. The limitations that exist in both measures are addressed using DL techniques that allow developing features using automated and data-driven methods. CNN models have achieved high diagnostic accuracy for AD diagnosis using MRI images especially in the discrimination of AD patients and cognitively normal subjects [2]. However, fine-grained classification tasks, including distinguishing between early mild impairment of the condition, the mild (MCI) and late MCI in multimedia unimodal MRI-only systems often fail because of the similarity of structural responses. Therefore, constructing multimodal fusion methods that merge imaging, cognitive and clinical data is a method that is progressively applied by researchers and these methods promise to improve diagnostic sensitivity and strength [3]. Notwithstanding such developments, multimodal systems introduce new challenges such as variance of data, increased architectural complexity, inability to interpret, and reduced generalizability across cohorts. Such considerations spur an extensive and pertinent analysis of research that has been published.

## **2.3 Definition**

Deep learning (DL) is a subarea of machine learning that utilizes multilayered neural networks which can directly learn from raw data hierarchies. The major application of DL in Alzheimer research is in the form of CNNs, attention-based networks and imaging and clinical data analysis hybrid architectures [1]. Multimodal Learning refers to an approach to using heterogeneous data (i.e., MRI images and cognitive scores) to generate the same predictive framework. Multimodal learning aims at capitalizing on the complementary information that cannot be realized on a single modality [3]. Attention mechanisms are neural elements, which dynamically emphasize salient characteristics or modalities during training thus boosting model execution and providing restricted interpretability by illuminating diagnostically significant data [4].

## **2.4 Detailed Literature Review**

Below is the detailed literature review.

## **2.4.1 Related Research Work 1: MRI-Based Deep Learning**

### **Approaches**

A number of publications that have been released in connection with the application of structural MRIs for Alzheimer's disease (AD) classification after 2020 highlight DL-based methods. Mohammed et al. (2024) review the current state of DL models for MRI classification and find that CNNs significantly surpass conventional classifiers in terms of AD structure classification[1]. The authors, nevertheless, stress that the majority of MRI-only systems are based on large labelled datasets and are characterized by a reduced generalization when imposed on unseen cohorts. A CNN -based MRI classification model suggested by Sorour et al. (2024) demonstrated excellent results in the initial AD detection, thus, proving the effectiveness of automated feature extraction [2]. Although these results are encouraging, the study is not expanded following imaging data and lacks the input of cognitive or the clinical data to draw conclusions on the applicability of the findings to a real-life diagnostic situation. On the same note, Raj et al. (2025) proposed a hybrid CNN that includes the InceptionV3 and ResNet-50 in the classification of Alzheimer. Although the model achieved high accuracy, it was more difficult to display the middle diseases stage which highlights the inherent weakness of unimodal imaging techniques in any fine-grained classification [3]. Taken together, these studies demonstrate MRI-based DL models are potent structural pattern recognition tools with a limitation in the form of the inability to combine both functional and clinical aspects of the Alzheimer disease.

## **2.4.2 Related Research Work 2: Multimodal Deep Learning**

### **Frameworks**

In order to overcome the drawbacks of unimodal systems, recent studies are more likely to use multimodal DL methodologies. Golovanevsky et al. (2022) proposed an attention-based multipurpose model that combines MRI and clinical data, which has better multi-class classification accuracy and indicates the advantage of cross-modal interaction [4]. However, their method requires careful alignment of features and is overwhelmed by greater computing complexity. Amine (2025) performed a review of multimodal DL models to detect early-stage Alzheimer, and found that the combination of imaging and cognitive information is always improved as compared to single-modality methods in the diagnosis of Alzheimer in its domains of early susceptibility.

The review, however, points out that most multimodal systems are based on the wider use of preprocessing and feature engineering, which suggests issues of scalability and reproducibility. Wang et al. (2024) came up with the concept of a hybrid multimodal model involving the integration of MRI and cognitive tests to forecast the clinical development of dementia. Though the model has been effective in illustrating the relationship between structural changes and cognitive decline, it was evaluated using one dataset alone thus restraining conclusions on matters of generalization [5]. Their multimodal neural network incorporated MRI, PET and neuropsychological scores, creating a lightweight, efficient neural network with competitive results (Gupta et al., 2025). Nonetheless, the practical applicability is reduced based on dependence on PET imaging because of its high cost and unavailability in standard clinical practice [6]. According to these studies, the concept of multimodal learning enhances the accuracy of the diagnosis and staging of the disease.

### **2.4.3 Multimodal Interpretability and Sophisticated Multimodal Models**

Recent sources emphasize the relevance of interpretability to the diagnostic systems based on the DL. One of the multimodal frameworks suggested by Castellano et al. (2024), which involves the use of both MRI and PET, has shown improved performance on detection of it but not regarding transparency of decisions made [7]. Raza et al. (2025) maintain that attention-based and explainable architectures are vital to clinical trust but cite that most of the existing models offer partial interpretability only [8]. Song et al. (2021) showed multimodal imaging fusion methods that retain modality-specific information and thus help to interpret the image but do not change performance [9]. Recent reviews incriminate transformer based multimodal architectures as a promise but an expensive and data intensive form of computation, which limits their utilization in resource constrained environments [10], [11].

## **2.5 Literature Review Summary Table**

Below is the literature review table.



**Table 2.1 Summary Table (Literature Review)**

| <b>No.</b> | <b>Study</b>               | <b>Modalities</b>     | <b>Method</b>          | <b>Strengths</b>          | <b>Limitations</b>    | <b>Accuracies</b> |
|------------|----------------------------|-----------------------|------------------------|---------------------------|-----------------------|-------------------|
| 1.         | Mohammed et al. (2024)     | MRI                   | DL Review              | Comprehensive analysis    | Unimodal focus        | 98.24%            |
| 2.         | Sorour et al. (2024)       | MRI                   | CNN                    | Strong early detection    | No clinical data      | 99.92%            |
| 3.         | Raj et al. (2025)          | MRI                   | Hybrid CNN             | High accuracy             | Poor stage separation | 96.6%             |
| 4.         | Golovanevsky et al. (2022) | MRI + Clinical        | Attention-based fusion | Improved multiclass       | High complexity       | 96.88%            |
| 5.         | Wang et al. (2024)         | MRI + Cognitive       | Hybrid DL              | Predicts progression      | Limited validation    | 90%               |
| 6.         | Gupta et al. (2025)        | MRI + PET + Cognitive | Lightweight DL         | Efficient fusion          | PET dependency        | 98.0%             |
| 7.         | Castellano et al. (2024)   | MRI + PET             | CNN                    | Enhanced accuracy         | Low interpretability  | 95.3%             |
| 8.         | Raza et al. (2025)         | Multimodal            | Review                 | Highlights explainability | Conceptual            | 92.14%            |
| 9.         | Song et al. (2021)         | MRI + PET             | Fusion DL              | Preserves modality info   | Limited staging       | 94.11%            |
| 10.        | Mohsen (2025)              | MRI                   | Review                 | Comparative insights      | Dataset bias          | 97.5%             |
| 11.        | Nagarajan et al. (2025)    | MRI                   | Survey                 | Comprehensive scope       | Lacks implementation  | 99.21%            |

## 2.6 Research Gap

The literature contains gaps. Even though there has been a lot of progress, some gaps still remain.

- Lack of interpretability of deep learning predictions to the clinical use.
- Weakness in extensive generalization of heterogeneous datasets.
- Algorithms on costly or unavailable modalities.
- Absence of end-to-end systems to integrate prediction and usable interfaces.

## 2.7 Problem Statement

Alzheimer's Disease (AD) is a progressive form of neurodegeneration and is one of the major causes of dementia in the world. This condition has great impacts on patients, their family members, and healthcare facilities. The manual examination of the imaging data makes it difficult to detect the MCI stage in AD because of the minute changes in the structure of the brain. The lack of access to specialists and modern technologies is a problem that leads to late and inaccurate diagnostics, especially in low-budget environments. MRI is one of the common methods used for assessing the disease. Still, there are some major problems with the current techniques. The structural changes in the brain in AD differ in size, location and magnitude between patients, making it difficult to accurately determine the disease-relevant areas. There is also inter-subject variability, differences in scanning protocols and also noise in MRI data which have negative impact on automated analysis. These factors make it difficult to develop discriminative features for reliable and accurate disease classification.

Due to their improved performance in classifying Alzheimer's disease, deep neural networks (DNNs) have found impressive success in a number of applications; however, there is a considerable gap between the beginning of their first and second symptoms. Some of the disadvantages are the training times, the computational expense of training, and a heavy reliance on huge, well-trained datasets.

In real world, these requirements might not be feasible. Models may perform poorly in complex situations and are prone to overfitting with small data sets. Therefore, it is difficult to deploy end-to-end deep learning models in real time or for large-scale applications. Clinical use still remains a major challenge.

There is an urgent requirement for a diagnostic framework for Alzheimer's disease, which should overcome the limitations through proper deep feature extraction, decreased training complexity, and improved generalization ability among other things. This research work intends to build a deep learning-based diagnostic model for Alzheimer's disease using pre-trained convolutional neural network for feature extraction from MRIs, then performing effective classification and multimodal fusion. The goal is to make a diagnosis of Alzheimer's disease possible at its early stages, avoiding the need for increased computation power for that purpose.

## **Chapter 3: Requirements and Design**

## Chapter 3: Requirements and Design

This chapter aims at clearly defining the needed system functionality and its structural and functional design in terms of making it reproducible by the readers with a suitable technical background. The chapter establishes functional and non-functional requirements, hardware and software dependencies, proposed methodological workflow, system architecture and exemplary use cases. It focuses on modularity, interpretability, and clinical relevance, as it is proposed to be done in medical artificial intelligence system design best practice.

### 3.1 Requirements

Below is the list of functional and non-functional requirements of our system.

#### 3.1.1 Functional Requirements

The following list identifies the functional requirements of the system's User.

Table 3.1.1.1: Functional Requirements of the User

**Table 3.1 Functional Requirements of User**

| ID     | Requirements                                                   |
|--------|----------------------------------------------------------------|
| FR-1.1 | The user should have the ability to create a profile.          |
| FR-1.2 | The user should have the ability to login.                     |
| FR-1.3 | The user should have the ability to upload MRI scan.           |
| FR-1.4 | The user should have the ability to perform a cognitive test.  |
| FR-1.5 | The user should have the ability to see the diagnostic report. |

3.1.1.2 Admin: In this section, the requirements that are functional in nature for the Admin role are described.

**Table 3.2 Functional Requirements of Admin**

| <b>ID</b> | <b>Requirements</b>                                                   |
|-----------|-----------------------------------------------------------------------|
| FR-2.1    | The admin should have the capability to log into the system.          |
| FR-2.2    | The admin should be able to see the list of all the registered users. |
| FR-2.3    | The admin should be able to control the user account.                 |
| FR-2.4    | The admin should be able to check the MRI and cognitive data.         |
| FR-2.5    | The admin should be able to update the model if required.             |

### **3.1.2 Non-Functional Requirements**

Here are the non-functional requirements of our system.

**Table 3.3 Non-Functional Requirements**

| <b>Non-Functional Requirement</b> | <b>Description</b>                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------|
| Performance Efficiency            | The system shall process individual MRI scans within an acceptable response time. |
| Usability                         | The system shall provide an intuitive user-friendly interface.                    |

### **3.1.3 Hardware and Software Requirements**

This system needs proper computational facilities to facilitate deep learning inference and processing data including:

- Multi-core CPU
- High end GPU
- 32GB RAM

These requirements are congruent to current research settings that are used in the development of AI in the medical profession.

## **3.2 Proposed Methodology**

Here we will discuss about the proposed methodology.

### **3.2.1 Overview**

Our system aims to detect and classify stages of Alzheimer's by using a web application integrated with deep learning model. Our system provides user friendly interface for users to verify the stage of Alzheimer effectively.

### **3.2.2 Steps**

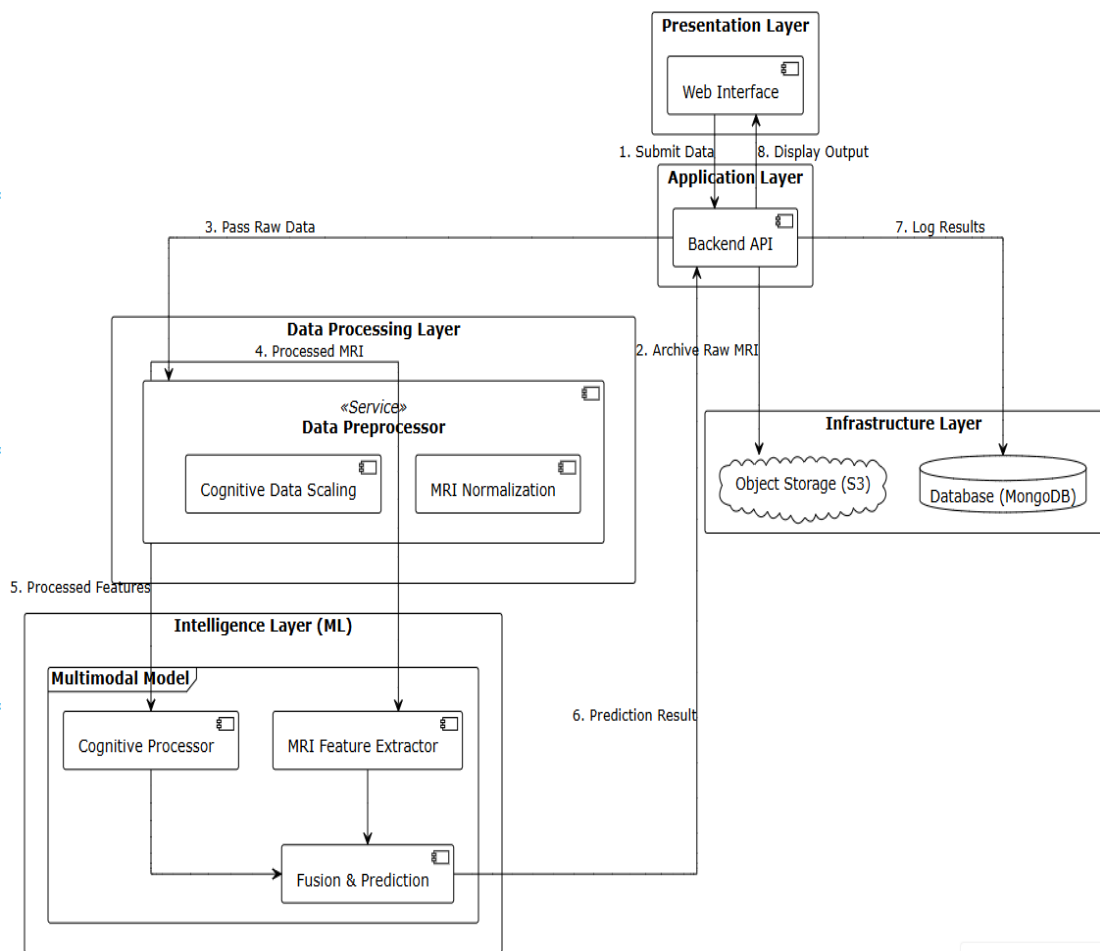
1. Data Collection: We collected the MRI dataset from ADNI which was not publicly available and was gotten upon request.
2. Data Preprocessing: Dataset was preprocessed from 3D to 2D with 35 center slices for each volume with dimensions 128x128. The dataset was split into train (70%), validation (20%), and test (10%).
3. Model Training and Evaluation: Train the CNN model using preprocessed dataset. The evaluation is done through confusion matrix.
4. Integration: Developing a web application which ensures user-friendly interface for users. Then we will integrate the model in our web app that will then do the detection based on the uploaded MRI and cognitive test taken and the system will generate the result on time.
5. Testing and Validation: Thorough testing will be done to ensure accuracy and reliability of our system and different CNN's will be trained to ensure we choose the best model.

## **3.3 System Architecture**

A modular, layered approach is used to develop the system architecture to help it to be scalable and maintainable. The architecture is as follows:

1. Data Input: The user uploads the MRI and takes cognitive test, which will work as input to system for classification.
2. Application Interface: The web application provides a user-friendly interface that allows easy submission of MRI images and cognitive test. It communicates with the model and ensures smooth interaction for user.

3. **Deep Learning Model:** The system uses a hybrid multimodal model that analyzes MRI images and cognitive tests. It processes the inputs and classifies the patient into different categories such as CN, EMCI, LMCI, or AD.
4. **Results:** The classification results are presented to the user, showing the predicted stage.



**Figure 3.1 Layered Architecture Diagram**



### 3.4 Use Cases

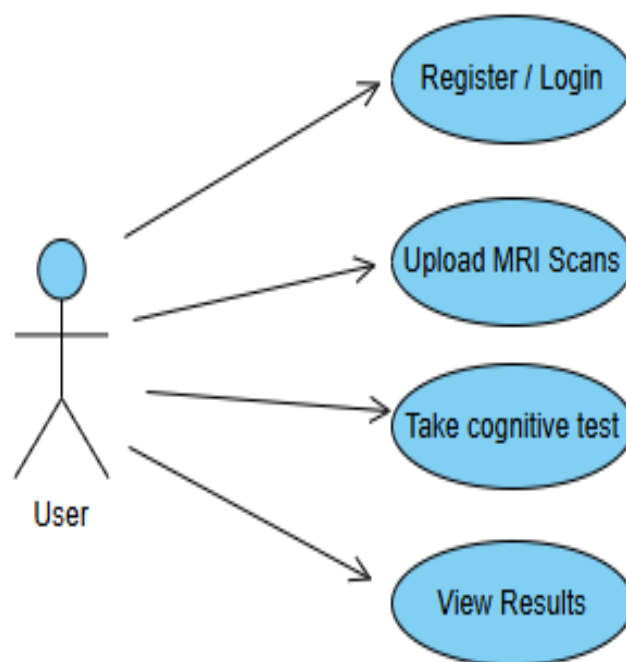
Below are the details of the use cases, complete use case and fully dressed use case.

#### 3.4.1 User

Below is the User use case.

**Table 3.4 Use Case Table (User)**

| Actors          | User                                                                                                                                                 |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-Conditions  | User must have MRI images ready<br>User must be registered in the system                                                                             |
| Main Flow       | User registers/logs-in to the system<br>User uploads MRI brain scans<br>User takes cognitive test<br>User views diagnostic result on their dashboard |
| Post-Conditions | Uploaded data is stored securely for future reference                                                                                                |



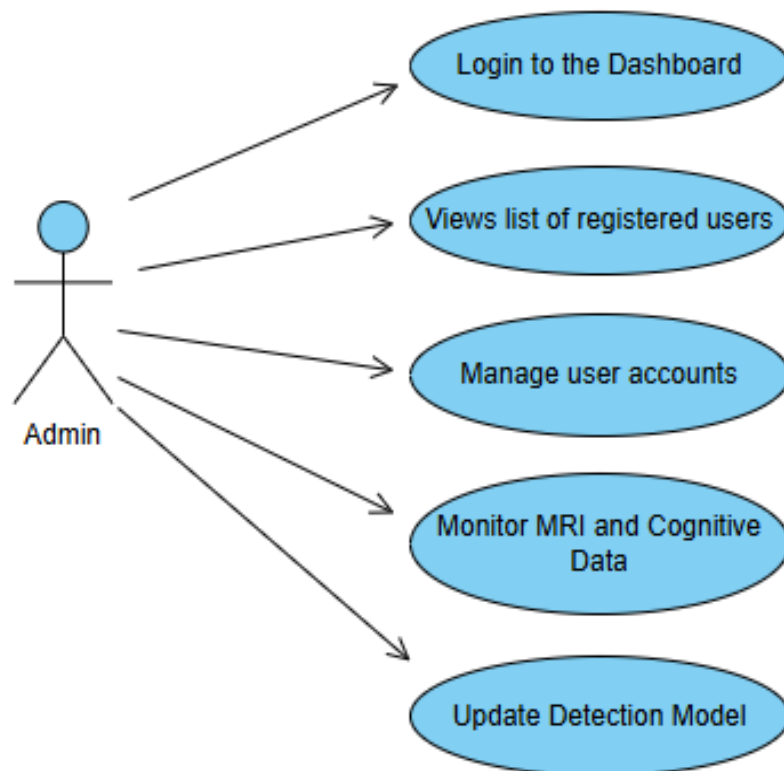
**Figure 3.2 User Use Case Diagram**

### 3.4.2 Admin

Below is the use case for Admin.

**Table 3.5 Use Case Table (Admin)**

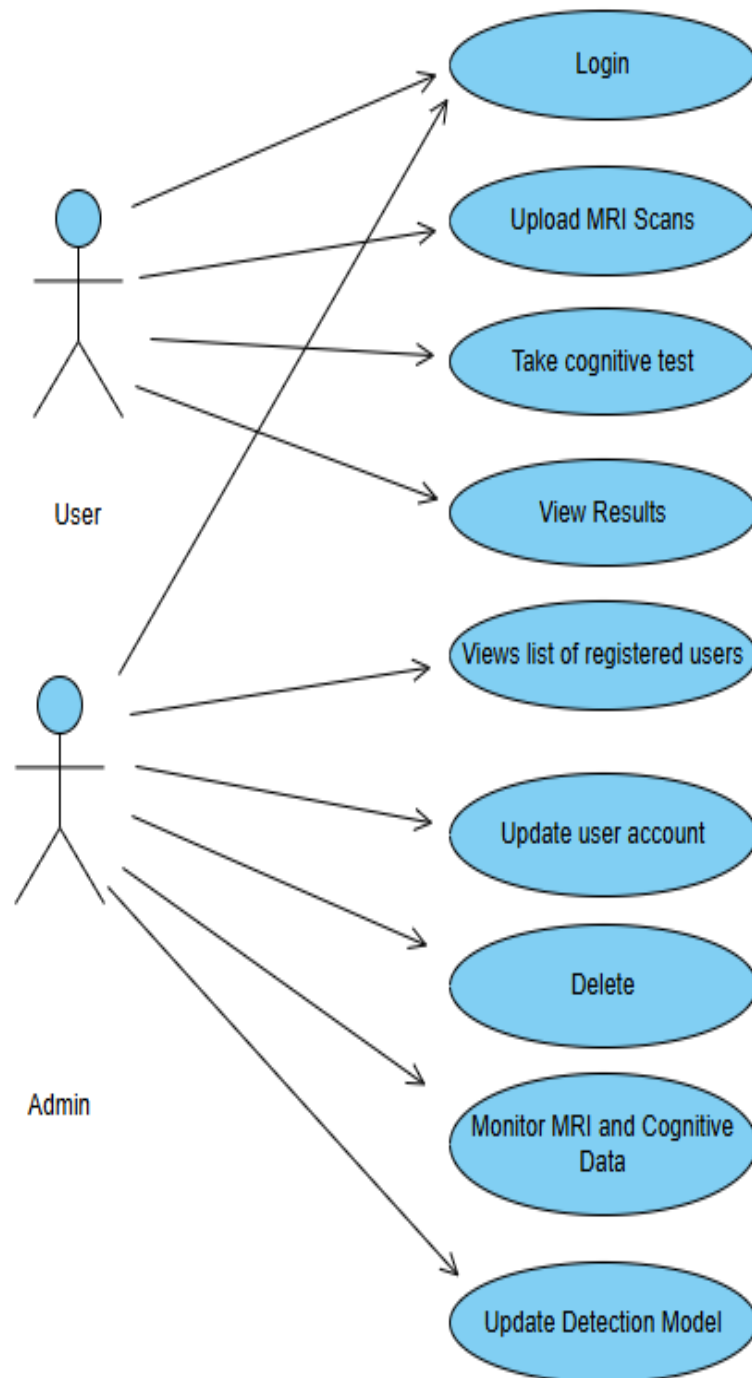
|                        |                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>          | <b>Admin</b>                                                                                                                                                                                                                 |
| <b>Pre-Conditions</b>  | Admin must have login credentials.                                                                                                                                                                                           |
| <b>Main Flow</b>       | Admin logs into the admin dashboard<br>Admin views the list of registered users<br>Admin can manage user accounts (update or delete account)<br>Admin can monitor MRI and cognitive data<br>Admin can update model if needed |
| <b>Post-Conditions</b> | Admin maintains system integrity                                                                                                                                                                                             |



**Figure 3.3 Admin Use Case Diagram**

### 3.4.3 Complete Use Case Diagram

Below is the complete Use Case Diagram.



**Figure 3.4 Complete Use Case Diagram**

### 3.4.4 Fully Dressed Use Case

Below are the fully dressed use cases.

#### 3.4.4.1 User Login

Below is the fully dressed use case for login.

**Table 3.6 User Login Fully Dressed Use Case Table**

|                        |                                                            |                                                                                               |                                                             |
|------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Name</b>            |                                                            | User Registration / Login                                                                     |                                                             |
| <b>Actor</b>           |                                                            | User                                                                                          |                                                             |
| <b>Summary</b>         |                                                            | Allows a new user to create an account or an existing user to access their private dashboard. |                                                             |
| <b>Pre-Conditions</b>  |                                                            | User has a stable internet connection and access to the web portal.                           |                                                             |
| <b>Post-Conditions</b> |                                                            | User is authenticated and redirected to their personal dashboard.                             |                                                             |
| <b>Basic Flow</b>      |                                                            |                                                                                               |                                                             |
| <b>Actor Action</b>    |                                                            | <b>System Response</b>                                                                        |                                                             |
| 1                      | User enters credentials (Email/Password) and clicks Login. | 2                                                                                             | System validates the credentials against the user database. |
|                        |                                                            | 3                                                                                             | System generates a secure session and grants access.        |
| 4                      | User views their personalized dashboard and past records.  |                                                                                               |                                                             |

### 3.4.4.2 Upload MRI

Below is the fully dressed use case of MRI Upload.

**Table 3.7 Upload MRI Fully Dressed Use Case Table**

|                        |                                                                |                        |                                                    |
|------------------------|----------------------------------------------------------------|------------------------|----------------------------------------------------|
| <b>Name</b>            | Upload MRI Scans                                               |                        |                                                    |
| <b>Actor</b>           | User                                                           |                        |                                                    |
| <b>Summary</b>         | Allows the user to upload MRI Scans.                           |                        |                                                    |
| <b>Pre-Conditions</b>  | User is logged in and has a valid MRI file.                    |                        |                                                    |
| <b>Post-Conditions</b> | The file is successfully stored in the backend for processing. |                        |                                                    |
| <b>Basic Flow</b>      |                                                                |                        |                                                    |
| <b>Actor Action</b>    |                                                                | <b>System Response</b> |                                                    |
| 1                      | User selects the MRI file and clicks the upload button.        | 2                      | System checks file format compatibility            |
|                        |                                                                | 3                      | System saves the file and confirms upload success. |
| 4                      | User confirms and proceeds to cognitive test.                  |                        |                                                    |

### 3.4.4.3 Take Cognitive Test

Below is the fully dressed use case of Cognitive Test.

**Table 3.8 Cognitive Test Fully Dressed Use Case Table**

|                        |                                            |                                                               |                                                        |
|------------------------|--------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------|
| <b>Name</b>            |                                            | Take Cognitive Test                                           |                                                        |
| <b>Actor</b>           |                                            | User                                                          |                                                        |
| <b>Summary</b>         |                                            | Allows the user to take cognitive test.                       |                                                        |
| <b>Pre-Conditions</b>  |                                            | User has uploaded the MRI successfully.                       |                                                        |
| <b>Post-Conditions</b> |                                            | Cognitive test has been done and ready for multimodal fusion. |                                                        |
| <b>Basic Flow</b>      |                                            |                                                               |                                                        |
| <b>Actor Action</b>    |                                            | <b>System Response</b>                                        |                                                        |
| 1                      | User answers the questions.                | 2                                                             | System validates that the scores are within the range. |
|                        |                                            | 3                                                             | System maps the score to a 30-bit binary vector        |
| 4                      | User proceeds to final diagnostic request. |                                                               |                                                        |

#### 3.4.4.4 View Diagnostic Result

Below is the fully dressed use case of View Diagnostic Result.

**Table 3.9 Diagnostic Results Fully Dressed Use Case Table**

|                 |                                                                                               |                 |                                                    |
|-----------------|-----------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------|
| Name            | View Diagnostic Result                                                                        |                 |                                                    |
| Actor           | User                                                                                          |                 |                                                    |
| Summary         | Allows the user to see the AI's classification based on the fusion of MRI and cognitive data. |                 |                                                    |
| Pre-Conditions  | Both MRI and Cognitive data have been submitted and processed.                                |                 |                                                    |
| Post-Conditions | User receives a diagnostic forecast (e.g., AD, CN, EMCI, or LMCI).                            |                 |                                                    |
| Basic Flow      |                                                                                               |                 |                                                    |
| Actor Action    |                                                                                               | System Response |                                                    |
| 1               | User clicks the "Generate Result" button on the dashboard.                                    | 2               | System runs the multimodal fusion model inference. |
|                 |                                                                                               | 3               | System displays the diagnostic label.              |
| 4               | User reads the detailed result on their dashboard.                                            |                 |                                                    |

### 3.4.4.5 Admin Login

Below is the fully dressed use case of Admin Login.

**Table 3.10 Admin Login Fully Dressed Use Case Table**

|                 |                                                                               |                 |                                                          |
|-----------------|-------------------------------------------------------------------------------|-----------------|----------------------------------------------------------|
| Name            | Admin Login                                                                   |                 |                                                          |
| Actor           | Admin                                                                         |                 |                                                          |
| Summary         | Provides the administrator with secure access to the system management tools. |                 |                                                          |
| Pre-Conditions  | Admin account exists with administrative privileges.                          |                 |                                                          |
| Post-Conditions | Admin is redirected to the management panel.                                  |                 |                                                          |
| Basic Flow      |                                                                               |                 |                                                          |
| Actor Action    |                                                                               | System Response |                                                          |
| 1               | Admin enters administrative credentials and clicks Login.                     | 2               | System verifies admin status and validates credentials.  |
|                 |                                                                               | 3               | System provides access to user lists and model controls. |
| 4               | Admin views the system health and user overview.                              |                 |                                                          |



### 3.4.4.6 View Registered Users

Below is the fully dressed use case of View Registered Users.

**Table 3.11 View Users Fully Dressed Use Case Table**

|                 |                                                              |                                                                                           |                                                            |
|-----------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Name            |                                                              | View Registered Users                                                                     |                                                            |
| Actor           |                                                              | Admin                                                                                     |                                                            |
| Summary         |                                                              | Allows the admin to oversee all active accounts within the Alzheimer's Diagnostic System. |                                                            |
| Pre-Conditions  |                                                              | Admin is successfully logged in to the admin dashboard.                                   |                                                            |
| Post-Conditions |                                                              | Admin can see a comprehensive list of all users and their status.                         |                                                            |
| Basic Flow      |                                                              |                                                                                           |                                                            |
| Actor Action    |                                                              | System Response                                                                           |                                                            |
| 1               | Admin clicks on the "User Management" tab.                   | 2                                                                                         | System fetches all user records from the primary database. |
|                 |                                                              | 3                                                                                         | System renders the list with search and filter options.    |
| 4               | Admin identifies the users currently utilizing the platform. |                                                                                           |                                                            |

### 3.4.4.7 Manage User Accounts

Below is the fully dressed use case of Manage User Accounts.

**Table 3.12 Manage Users Fully Dressed Use Case Table**

|                        |                                                             |                                                                   |                                                                 |
|------------------------|-------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Name</b>            |                                                             | Manage User Accounts                                              |                                                                 |
| <b>Actor</b>           |                                                             | Admin                                                             |                                                                 |
| <b>Summary</b>         |                                                             | Allows the admin to update or delete users.                       |                                                                 |
| <b>Pre-Conditions</b>  |                                                             | Admin is viewing the list of registered users.                    |                                                                 |
| <b>Post-Conditions</b> |                                                             | User records are modified or permanently removed from the system. |                                                                 |
| <b>Basic Flow</b>      |                                                             |                                                                   |                                                                 |
| <b>Actor Action</b>    |                                                             | <b>System Response</b>                                            |                                                                 |
| 1                      | Admin selects a user and clicks "Update" or "Delete".       | 2                                                                 | System prompts for a second confirmation to prevent error.      |
|                        |                                                             | 3                                                                 | System executes the database query to modify/remove the record. |
| 4                      | Admin receives a notification that the change is finalized. |                                                                   |                                                                 |

### 3.4.4.8 Monitor MRI and Cognitive Data

Below is the fully dressed use case of Monitor MRI and Cognitive Data.

**Table 3.13 Monitor MRI and Cog Data Fully Dressed Use Case Table**

|                        |                                                                 |                                                                                    |                                                               |
|------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Name</b>            |                                                                 | Monitor MRI and Cognitive Data                                                     |                                                               |
| <b>Actor</b>           |                                                                 | Admin                                                                              |                                                               |
| <b>Summary</b>         |                                                                 | Allows the admin to monitor the type and volume of data being processed by the AI. |                                                               |
| <b>Pre-Conditions</b>  |                                                                 | Admin is logged in and system data is being actively collected.                    |                                                               |
| <b>Post-Conditions</b> |                                                                 | Admin can ensure that data heterogeneity and imbalances are tracked.               |                                                               |
|                        |                                                                 |                                                                                    |                                                               |
| <b>Basic Flow</b>      |                                                                 |                                                                                    |                                                               |
| <b>Actor Action</b>    |                                                                 | <b>System Response</b>                                                             |                                                               |
| 1                      | Admin navigates to the "Data Analytics" section.                | 2                                                                                  | System displays statistics on uploaded scans and test scores. |
|                        |                                                                 | 3                                                                                  | System highlights any anomalies or data processing errors.    |
| 4                      | Admin ensures the system is handling data partitions correctly. |                                                                                    |                                                               |

### 3.4.4.9 Update Model

Below is the fully dressed use case of Update Model.

**Table 3.14 Update Model Fully Dressed Use Case Table**

|                        |                                                                                                    |                        |                                                            |
|------------------------|----------------------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------|
| <b>Name</b>            | Update Model                                                                                       |                        |                                                            |
| <b>Actor</b>           | Admin                                                                                              |                        |                                                            |
| <b>Summary</b>         | Allows the admin to replace the current AI model with a newly trained version to improve accuracy. |                        |                                                            |
| <b>Pre-Conditions</b>  | Admin has a new validated weights file (.h5) ready for deployment.                                 |                        |                                                            |
| <b>Post-Conditions</b> | The backend begins using the updated model for all new user requests.                              |                        |                                                            |
| <b>Basic Flow</b>      |                                                                                                    |                        |                                                            |
| <b>Actor Action</b>    |                                                                                                    | <b>System Response</b> |                                                            |
| 1                      | Admin uploads the new model file in the settings panel.                                            | 2                      | System verifies the file structure and model architecture. |
|                        |                                                                                                    | 3                      | System hot-swaps the model files in the backend directory. |
| 4                      | Admin receives a success message indicating the new model is live.                                 |                        |                                                            |

## **Chapter 4: Implementation and Test Cases**

## Chapter 4: Implementation and Case Studies

This chapter highlights the specific design choices made during the development of the AD detection system and describes the architecture of the prototype so that others can replicate the process and results.

### 4.1 Implementation

This section details how the multimodal framework was converted into a functional software system. The implementation is based on the smooth integration between the deep learning backend and a modern web front-end.

#### 4.1.1 Development Stack and Technical Environment

To ensure the system is both powerful and accessible, we utilized a decoupled architecture:

1. Backend & AI: Node.js was used for backend, model was developed in Python using TensorFlow and Keras for execution. We used FastAPI to handle requests between the AI and the web interface.
2. Frontend Dashboard: React.js was used for frontend development providing a highly responsive and intuitive user interface.

#### 4.1.2 Imaging and Preprocessing Pipeline

The imaging pipeline ensures that raw brain scans are compatible with the neural network's input shape.

1. Data Handling: The system communicates with 3D NIFTI volumes via the Nibabel library. It detects and extracts the main axial slices, which contain the most important indicators of hippocampal shrinkage.
2. Normalization: We implemented Z-score Intensity Normalization and resized the slices to a uniform 128x128 pixels. This will ensure that differences in MRI are kept to a minimum scanner hardware and do not affect the model's accuracy.
3. Hybrid Feature Extraction: MRI branch performs a dual-stream feature extraction ensemble. InceptionV3 captures multi-scale spatial details and MobileNet offers efficient and high-level structural patterns that ensure the model is easy to deploy in the web with a lightweight footprint.

### 4.1.3 Cognitive Vectorization and Fusion Logic

To incorporate clinical data, I developed a custom feature-engineering pipeline for the cognitive assessments:

1. **30-Question Vector:** The model does not get raw scores from MoCA, ADAS-13, and CDR-SB. Thus, I created a logic-gate function that converts these scores into a binary vector with 30 bits. Memory, orientation, and executive function are three separate cognitive domains that are represented as active or inactive bits.
2. **Attention-Based Fusion:** When combined with backbone attentive fusion, the model can focus on specific areas of the image instead of the entire structure. Attention weights are determined by the system for that specific patient once the MRI features and cognitive vector converge at the fusion head to prioritize specific data, leading to the high accuracy for the final testing phase.
3. **Web Integration and Interface Workflow:** The React dashboard design guarantees that non-technical users can access the complex backend.
4. **Asynchronous Prediction:** The React frontend sends an asynchronous request when the user uploads an MRI file and inputs cognitive scores.
5. **Inference Management:** The trained weights are loaded in the backend (incv3\_mobnet\_cog30\_fusion\_model.h5). It makes the prediction across the four classes (AD, CN, EMCI, and LMCI).
6. **Result Delivery:** The prediction is given back to the dashboard within 1 to 5 seconds. The interface then displays the diagnostic forecast alongside the classification label, completing the end-to-end diagnostic support loop.

## 4.2 Test case Design and description

Here, we have given some test cases to check if the Alzheimer's Detection system is working fine or not. These tests were designed to see how well the system coped with brain scans and cognitive test scores at the same time. We have tested all the main points, from uploading the data and processing the images to the final prediction and how the user interface looks. We used standard testing methods to make sure that the system is not only accurate in theory but also stable and easy to use for anyone trying it.

### 4.2.1 Test Case 1: User Registration and Authentication

Below is the test case of User Registration and Authentication.

**Table 4.1 Test Case Table (User Registration)**

| Authentication Module                                                                                              |                                                                               |                                                    |                        |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------|------------------------|
| UC-01 User Login/Registration                                                                                      |                                                                               |                                                    |                        |
| Test Case ID:                                                                                                      | TC-01                                                                         | Test Date:                                         | 04/04/2026             |
| Test case Version:                                                                                                 | 1.0                                                                           | Use Case Reference(s):                             | User Registers/Logs-in |
| Revision History:                                                                                                  | New test case                                                                 |                                                    |                        |
| Objective                                                                                                          | Verify that a new user can register and an existing user can log in securely. |                                                    |                        |
| Product/Ver/Module:                                                                                                | Alzheimer's Disease Detection / v1.0 / Auth Module                            |                                                    |                        |
| Environment:                                                                                                       | Web browser, Stable Internet, Backend Server running                          |                                                    |                        |
| Assumptions:                                                                                                       | Database is connected and email unique constraints are active.                |                                                    |                        |
| Pre-Requisite:                                                                                                     | User must have a valid email address not previously registered.               |                                                    |                        |
| Step No.                                                                                                           | Execution description                                                         | Procedure result                                   |                        |
| 1                                                                                                                  | Navigate to signup page and enter details.                                    | System creates account and redirects to dashboard. |                        |
| 2                                                                                                                  | Logout and attempt to log back in with credentials.                           | System validates credentials and grants access.    |                        |
| Comments:                                                                                                          |                                                                               |                                                    |                        |
| Ensure Password Hashing is implemented for security.                                                               |                                                                               |                                                    |                        |
| ✓ <input checked="" type="checkbox"/> Passed <input type="checkbox"/> Failed <input type="checkbox"/> Not Executed |                                                                               |                                                    |                        |



## 4.2.2 Test Case 2: Multimodal Data Upload

Below is the test case of Multimodal Data Upload.

**Table 4.2 Test Case Table (Multimodal Data Upload)**

| MRI Upload and Cognitive Test                                                                                      |                                                                               |                                                    |                                  |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------|
| UC-02 and UC-03 Data Submission                                                                                    |                                                                               |                                                    |                                  |
| Test Case ID:                                                                                                      | TC-02                                                                         | Test Date:                                         | 06/04/2026                       |
| Test case Version:                                                                                                 | 1.0                                                                           | Use Case Reference(s):                             | Upload MRI / Take Cognitive Test |
| Revision History:                                                                                                  | New test case                                                                 |                                                    |                                  |
| Objective                                                                                                          | Verify that a new user can register and an existing user can log in securely. |                                                    |                                  |
| Product/Ver/Module :                                                                                               | Alzheimer’s Disease Detection / v1.0 / Auth Module                            |                                                    |                                  |
| Environment:                                                                                                       | Web browser, Stable Internet, Backend Server running                          |                                                    |                                  |
| Assumptions:                                                                                                       | Database is connected and email unique constraints are active.                |                                                    |                                  |
| Pre-Requisite:                                                                                                     | User must have a valid email address not previously registered.               |                                                    |                                  |
| Step No.                                                                                                           | Execution description                                                         | Procedure result                                   |                                  |
| 1                                                                                                                  | Navigate to signup page and enter details.                                    | System creates account and redirects to dashboard. |                                  |
| 2                                                                                                                  | Logout and attempt to log back in with credentials.                           | System validates credentials and grants access.    |                                  |
| Comments:                                                                                                          |                                                                               |                                                    |                                  |
| Validation needed to ensure only image files are uploaded.                                                         |                                                                               |                                                    |                                  |
| ✓ <input checked="" type="checkbox"/> Passed <input type="checkbox"/> Failed <input type="checkbox"/> Not Executed |                                                                               |                                                    |                                  |

### 4.2.3 Test Case 3: Result Display

Below is the Test Case 3 of Result Display.

**Table 4.3 Test Case Table (Result Display)**

| ML Prediction Engine                                                                                               |                                                                      |                                              |                              |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------|------------------------------|
| UC-04 View Results                                                                                                 |                                                                      |                                              |                              |
| Test Case ID:                                                                                                      | TC-03                                                                | Test Date:                                   | 06/04/2026                   |
| Test case Version:                                                                                                 | 1.0                                                                  | Use Case Reference(s):                       | User views diagnostic result |
| Revision History:                                                                                                  | Testing the model output                                             |                                              |                              |
| Objective                                                                                                          | Verify the system displays the correct classification based on data. |                                              |                              |
| Product/Ver/Module:                                                                                                | Alzheimer's Disease Detection / v1.0 / Result                        |                                              |                              |
| Environment:                                                                                                       | Server with Tensorflow/Keras environment loaded.                     |                                              |                              |
| Assumptions:                                                                                                       | The model weights are correctly loaded in the backend.               |                                              |                              |
| Pre-Requisite:                                                                                                     | MRI and cognitive data must be submitted for the current session.    |                                              |                              |
| Step No.                                                                                                           | Execution description                                                | Procedure result                             |                              |
| 1                                                                                                                  | Click "Generate Report" after data submission.                       | Model processes data and returns a category. |                              |
| 2                                                                                                                  | Check user dashboard for the result display.                         | Result is visible.                           |                              |
| Comments:                                                                                                          |                                                                      |                                              |                              |
| The fusion logic must correctly weight test score.                                                                 |                                                                      |                                              |                              |
| ✓ <input checked="" type="checkbox"/> Passed <input type="checkbox"/> Failed <input type="checkbox"/> Not Executed |                                                                      |                                              |                              |

## 4.2.4 Test Case 4: Admin User Management

Below is the Test Case 4 of Admin User Management.

**Table 4.4 Test Case Table (Admin Management)**

| Admin Dashboard                                                                                                    |                                                                          |                                                      |                                |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------|--------------------------------|
| UC - Admin Manage                                                                                                  |                                                                          |                                                      |                                |
| Test Case ID:                                                                                                      | TC-04                                                                    | Test Date:                                           | 10/04/2026                     |
| Test case Version:                                                                                                 | 1.0                                                                      | Use Case Reference(s):                               | Manage accounts / Monitor Data |
| Revision History:                                                                                                  | Testing Administrative CRUD operations                                   |                                                      |                                |
| Objective                                                                                                          | Ensure Admin can view, update, and delete user profile and monitor data. |                                                      |                                |
| Product/Ver/Module:                                                                                                | Alzheimer's Disease Detection / v1.0 / Admin                             |                                                      |                                |
| Environment:                                                                                                       | Admin level credentials and populated database                           |                                                      |                                |
| Assumptions:                                                                                                       | Admin has higher privilege levels than users.                            |                                                      |                                |
| Pre-Requisite:                                                                                                     | At least one user is registered in the system.                           |                                                      |                                |
| Step No.                                                                                                           | Execution description                                                    | Procedure result                                     |                                |
| 1                                                                                                                  | Login as Admin and view User list.                                       | All registered users are displayed correctly.        |                                |
| 2                                                                                                                  | Select a user and click Delete/Update.                                   | Database is updated and user is removed or modified. |                                |
| 3                                                                                                                  | Access the data monitor to view MRI/Cog result.                          | Admin can see data log for system auditing.          |                                |
| Comments:                                                                                                          |                                                                          |                                                      |                                |
| Crucial for Data protection compliance.                                                                            |                                                                          |                                                      |                                |
| ✓ <input checked="" type="checkbox"/> Passed <input type="checkbox"/> Failed <input type="checkbox"/> Not Executed |                                                                          |                                                      |                                |

## 4.3 Test Metric

Test metrics provide quantitative measures for assessing the effectiveness and reliability of test cases. Even at the prototype stage, defining evaluation metrics ensures that system validation follows a disciplined and transparent approach.

### 4.3.1 Test Case Matrix

Below is the test case matrix.

**Table 4.5 Test Matrix**

| <b>Metric:</b>                      | <b>Purpose</b>                                                                                                                                                                                           |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Number of Test Cases:</b>        | 4                                                                                                                                                                                                        |
| <b>Number of Test Cases Passed:</b> | 4                                                                                                                                                                                                        |
| <b>Number of Test Cases Failed:</b> | 0                                                                                                                                                                                                        |
| <b>Test Case Defect Density:</b>    | $(0/4) * 100 = 0\%$                                                                                                                                                                                      |
| <b>Test Case Effectiveness:</b>     | 100%                                                                                                                                                                                                     |
| <b>Traceability Matrix:</b>         | Traceability is maintained by mapping each functional requirement (Login, MRI Upload, Cognitive Test, and Admin Control) to its corresponding test case to ensure all implemented features are verified. |

## **Chapter 5: Experimental Results and Analysis**

## **Chapter 5: Experimental Results and Analysis**

This chapter contains experimental setup details, performance and results of different model, model comparison and conclusion.

### **5.1 Introduction**

This chapter presents the experimental findings of our Alzheimer’s disease detection system. We evaluate the performance of our proposed multimodal fusion approach, which combines structural brain data (MRI) with functional cognitive assessments. The primary goal is to demonstrate that integrating these two distinct types of data leads to several architectures, including ResNet50, DenseNet, VGG16, and our final hybrid multimodal model.

### **5.2 Experimental Setup**

All experiments were performed under a standardized environment.

1. **Software & Frameworks:** The system was built with Python 3.12, using TensorFlow and Keras for deep learning, and Scikit-Learn for statistical evaluation.
2. **Hardware:** Training was done on Google Colab, using high-performance GPUs to handle the heavy computational load of image processing.
3. **Preprocessing:** All raw NIfTI scans were normalized using Z-score scaling and resized to a 128 x 128 resolution. We extracted the middle 35 axial slices from each brain volume to focus on the areas that are most affected by atrophy like hippocampus.
4. **Data Split:** The dataset was split into a standard split: 70% training, 20% validation, and 10% testing, ensuring that each stage had an equal representation of AD, CN, EMCI, and LMCI cases.

### **5.3 Performance Result and Analysis**

This section details the performance of different models tested during the development phase.

### 5.3.1 VGG16\_ LeakyReLU

The suggested network architecture is based on the modified VGG16 architecture, with an addition of Attention Pooling layer along with the LeakyReLU activation function, to categorize the MRI scans into four different classes including AD, CN, LMCI, and EMCI. The first ten layers were kept frozen while others were fine-tuned. The network uses a combination of Attention Pooling, which adaptively weights the spatial features and focuses on the high-relevance regions of the MRI slices, and LeakyReLU activation ( $\alpha = 0.1$ ) to mitigate the Dying ReLU problem in the fine-tuning stage.

#### 5.3.1.1 Confusion Matrix

Below is the Confusion Matrix of VGG16\_LeakyReLU.

**Table 5.1 Confusion Matrix VGG16\_LeakyReLU**

|      | AD   | CN   | LMCI | EMCI |
|------|------|------|------|------|
| AD   | 1597 | 37   | 56   | 25   |
| CN   | 16   | 4876 | 85   | 63   |
| LMCI | 46   | 167  | 4971 | 101  |
| EMCI | 24   | 60   | 72   | 3134 |

#### 5.3.1.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.2 Evaluation Matric VGG16\_LeakyReLU**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.9510    | 0.9509 | 0.9509   | 0.9509   |

### 5.3.2 ResNet50\_LeakyReLU

The model used in this study is based on the ResNet50 architecture, a deep residual network that enables training of very deep systems using skip connections. Differing from the VGG16 approach, this implementation utilizes a Global Average Pooling (GAP) layer after the convolutional base to reduce the feature dimension while

preserving the spatial information. To adapt the pre-trained model to the MRI dataset, we employed a selective fine-tuning strategy by unfreezing the last 30 layers. This enabled the model to modify its high-level abstract filters to the specific structure of the Alzheimer’s disease progression while keeping the robust low-level features learned from the ImageNet dataset.

### 5.3.2.1 Confusion Matrix

Below is the Confusion Matrix of ResNet50\_LeakyReLU.

**Table 5.3 Confusion Matrix ResNet50\_LeakyReLU**

|      | AD   | CN   | LMCI | EMCI |
|------|------|------|------|------|
| AD   | 1512 | 63   | 61   | 79   |
| CN   | 61   | 4697 | 170  | 112  |
| LMCI | 86   | 400  | 4587 | 212  |
| EMCI | 55   | 66   | 129  | 3040 |

### 5.3.2.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.4 Evaluation Matric ResNet50\_LeakyReLU**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.9033    | 0.9025 | 0.9023   | 0.9025   |

### 5.3.3 IncV3\_MobNet\_LeakyReLU

We evaluated a model which is a Hybrid Ensemble Architecture that combines the ability of InceptionV3 to extract multi-scale features with the low computational cost and depth-wise separate convolutions of MobileNet. The dual-stream model was implemented to capture the complex structural patterns and specific variations in intensity from the MRI scans. The outputs from both the pre-trained backbones were fused through a Concatenation layer which produced a high dimensional feature vector



that was then processed by a dense classification head. To avoid the increased risk of over-fitting caused by such a large parameter space, the model used L2 Regularization (0.001) and a large Dropout rate (0.4).

#### 5.3.3.1 Confusion Matrix

Below is the Confusion Matrix of IncV3\_MobNet\_LeakyReLU.

**Table 5.5 Confusion Matrix IncV3\_MobNet\_LeakyReLU**

|      | AD   | CN   | LMCI | EMCI |
|------|------|------|------|------|
| AD   | 1390 | 131  | 94   | 100  |
| CN   | 41   | 4557 | 228  | 214  |
| LMCI | 86   | 632  | 4415 | 152  |
| EMCI | 33   | 223  | 182  | 2852 |

#### 5.3.3.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.6 Evaluation Matric IncV3\_MobNet\_LeakyReLU**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.8645    | 0.8619 | 0.8619   | 0.8619   |

#### 5.3.4 IncV3\_DenseNet121

This experiment is a Custom Hybrid Architecture that combines the two powerful models in deep learning, the multi-scale feature extraction of Inception and the feature-reuse mechanism of DenseNet. Instead of standard pre-trained backbones like our earlier tests, this model uses a custom lightweight Inception-V3 base and a DenseNet-121 structure fine-tuned for 128x128 grayscale MRI input.

The Inception branch uses different convolutional paths to capture spatial hierarchies. Meanwhile, the DenseNet branch is made up of a sequence of dense blocks with a

growth rate of 16, keeping the flow of the gradient smooth through feature concatenation.

#### 5.3.4.1 Confusion Matrix

Below is the Confusion Matrix of IncV3\_DenseNet121.

**Table 5.7 Confusion Matrix IncV3\_DenseNet121**

|      | AD  | CN   | LMCI | EMCI |
|------|-----|------|------|------|
| AD   | 915 | 208  | 583  | 9    |
| CN   | 15  | 3552 | 1462 | 11   |
| LMCI | 3   | 158  | 5105 | 19   |
| EMCI | 48  | 416  | 19   | 991  |

#### 5.3.4.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.8 Evaluation Matric IncV3\_DenseNet121**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.7761    | 0.6890 | 0.6702   | 0.6890   |

#### 5.3.5 ResNet50\_Cog

This model uses high level spatial features from a ResNet50 backbone with a 30 dimensional feature vector from scores in a Cognitive Assessment. The implementation is a critical component and is shown in the Patient ID Alignment phase with a numeric matching logic to align clinical records with their matching MRI scans using the RID identifier. Technically, the model uses a Dual-Branch Attention Mechanism, in which a Softmax-based attention head calculates specific weights for each modality. This allows the system to automatically pick the most distinguishing data either from neuroimaging features or from cognitive markers depending on the input of a given stream for a given diagnostic class. The model proves to be of high confidence in

classifying AD, CN, EMCI and LMCI with an accuracy of 97.24% and balanced F1-Score of 97.23%. The attention aware integration of clinical indicators proves to be effective in solving the diagnostic issues and creating a robust system.

#### 5.3.5.1 Confusion Matrix

Below is the Confusion Matrix of ResNet50\_Cog.

**Table 5.9 Confusion Matrix ResNet50\_Cog**

|      | AD | CN  | LMCI | EMCI |
|------|----|-----|------|------|
| AD   | 46 | 1   | 1    | 1    |
| CN   | 0  | 128 | 1    | 0    |
| LMCI | 0  | 3   | 159  | 2    |
| EMCI | 0  | 0   | 3    | 89   |

#### 5.3.5.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.10 Evaluation Matric ResNet50\_Cog**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.9725    | 0.9723 | 0.9723   | 0.9723   |

#### 5.3.6 IncV3\_MobNet\_Attention\_Pooling

The experiment consists of implementing a complex Hybrid Dual-Stream Architecture, which combines the structural depth of InceptionV3 with the computational efficiency of MobileNet, and is further enhanced by a custom Attention Pooling mechanism. Instead of using the standard global pooling methods, Attention Pooling assigns weights to feature maps through a dense layer with softmax activation to dynamically learn spatial importance so that the system can focus on the most clinically relevant areas of the brain MRI. The dataset was in grayscale, so to solve the limitations, a channel-triplication approach was used. The top layers of both the backbones (layer 250+ for InceptionV3 and the last 25 layers for MobileNet) were carefully fine-tuned to improve the pre-trained features for medical diagnostic accuracy.

### 5.3.6.1 Confusion Matrix

Below is the Confusion Matrix of IncV3\_Mobnet\_Attention\_Pooling.

**Table 5.11 Confusion Matrix IncV3\_MobNet\_Att\_Pooling**

|      | AD   | CN   | LMCI | EMCI |
|------|------|------|------|------|
| AD   | 1577 | 42   | 43   | 53   |
| CN   | 21   | 4846 | 103  | 70   |
| LMCI | 47   | 255  | 4863 | 120  |
| EMCI | 24   | 54   | 95   | 3117 |

### 5.3.6.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.12 Evaluation Matric IncV3\_MobNet\_Att\_Pooling**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.9398    | 0.9395 | 0.9394   | 0.9395   |

### 5.3.7 IncV3\_MobNet\_Cog\_Fusion\_Softmax (Proposed Model)

We selected the InceptionV3-MobileNet-Cognitive Fusion model as the final proposed architecture because of its better performance and its advanced capability to simulate clinical diagnostic workflows by combining physiological and functional data streams. The model basically uses an ensemble feature extraction method that combines InceptionV3 multi-scale processing and MobileNet depth-wise convolutional efficiency to capture a broad range of neuroanatomical biomarkers.

The development of a Patient-Centric data alignment algorithm is a unique contribution of this work, which mathematically aligns spatial MRI features with a 30-dimensional cognitive profile, derived from MOCA, CDRSB and ADAS13 assessments using the RID identifier. This suggests that the model is working on a generic diagnostic profile for each subject.

The architecture proposes to combine these heterogeneous inputs through a Softmax based Attention Fusion mechanism that acts as a learnable keeper that automatically assigns the importance of either the imaging features or the clinical markers, depending on the discriminative power of these features for the scenario at the time. The model shows a high sensitivity to detect the transitional stages of neurodegeneration, especially EMCI and LMCI with a maximum accuracy of 98.84% and F1-score of 98.84%. This attention-aware framework is able to effectively resolve the visual uncertainty commonly found in unimodal systems, resulting in a robust and highly accurate support tool for automated Alzheimer’s diagnosis.

### 5.3.7.1 Confusion Matrix

Below is the Confusion Matrix of IncV3\_MobNet\_Cog\_Fusion\_Softmax.

**Table 5.13 Confusion Matrix IncV3\_MobNet\_Cog\_Fusion\_Softmax**

|      | AD | CN  | LMCI | EMCI |
|------|----|-----|------|------|
| AD   | 48 | 0   | 0    | 1    |
| CN   | 0  | 127 | 2    | 0    |
| LMCI | 0  | 0   | 164  | 0    |
| EMCI | 0  | 1   | 1    | 90   |

### 5.3.7.2 Performance Evaluation

Below are the evaluation metrics.

**Table 5.14 Evaluation Matric IncV3\_MobNet\_Cog\_Fusion\_Softmax**

| Precision | Recall | F1-Score | Accuracy |
|-----------|--------|----------|----------|
| 0.9887    | 0.9885 | 0.9884   | 0.9884   |

### 5.3.8 Model Comparison

Here is the comparison of all the models.

**Table 5.15 Model Comparison Table**

| <b>Model Architecture</b>                                 | <b>Accuracy</b> | <b>F1-Score</b> |
|-----------------------------------------------------------|-----------------|-----------------|
| VGG16_LeakyReLU                                           | 95.09%          | 0.9509          |
| ResNet50_LeakyReLU                                        | 90.25%          | 0.9023          |
| Inceptionv3_MobileNet_LeakyReLU                           | 86.19%          | 0.8619          |
| InceptionV3_DenseNet121_LR                                | 68.90%          | 0.6702          |
| ResNet50_Cog                                              | 97.23%          | 0.9723          |
| IncV3_MobNet_Attention_Pooling                            | 93.95%          | 0.9394          |
| Inceptionv3_MobNet_Cog_Fusion_Softmax<br>(Proposed Model) | 98.84%          | 0.9884          |

### 5.3.9 Detailed Analysis of Final Model

Our final model, which fuses Inceptionv3 and MobileNet features with a 30-question cognitive vector showed nearly perfect classification. According to confusion matrix:

1. AD & LMCI: These categories were identified with 100% precision, as the structural damage in late stages is highly distinct.
2. EMCI & CN: Only a handful of cases (approx. 2-3) were misclassified between "Normal" and "Early Mild Cognitive Impairment," which are the most difficult categories to distinguish.

## 5.4 Analysis and Discussion

The results provide several critical insights into the effectiveness of multimodal learning for Alzheimer's detection.

1. The Power of Fusion: The MRI-only model did have a good result (93.95%) but the addition of the cognitive vector brought the accuracy up to 98.85%. This

confirms that "functional" data (how a patient thinks) fills the gaps where "structural" data (how the brain looks) might be ambiguous.

2. Attention Mechanism: An attention fusion technique was utilized so that the model can recognize the importance of each feature for a given patient.
3. Overfitting Control: We observed that simpler fusion (like our ResNet50-MLP) was prone to overfitting. By introducing Dropout layers, L2 Regularization, and Batch Normalization in our final hybrid model, we achieved a stable validation curve and excellent generalizability.

## **5.5 Conclusion**

Finally, the experimental results successfully confirm our claim. The best performing variant was the Incv3\_Mobnet\_Cog\_Fusion model, which achieved an outstanding accuracy of 98.85%. We have developed a robust diagnostic tool that can precisely identify subtle stages of Alzheimer's disease by fusing high-level spatial features from two different image encoders and a binarized cognitive question vector. The model shows high potential as a clinical decision support tool for early intervention.

## **Chapter 6: Conclusion and Future Directions**



## **Chapter 6: Conclusion and Future Perspectives**

This chapter describes conclusions of the work and possible future perspectives on research along with the problems met during research.

### **6.1 Conclusion**

This paper introduces an automated AI-based system used for detection of Alzheimer's disease (AD) and its classification into four groups: AD, CN, EMCI and LMCI. The novelty of the project is in processing the complex 3D MRIs (magnetic resonance imaging) obtained from the ADNI dataset and converting them into the form usable for deep learning.

Multimodal fusion technique was implemented going beyond the use of only imaging data that is typical for traditional approaches. We introduced a specific "30-question cognitive vector" based on the results of clinical tests (MOCA, ADAS-C13, CDRSB) and added them to visual features extracted by a multimodal approach based on InceptionV3 and MobileNet models.

The major conclusion of the project is in the superiority of multimodal approaches compared with single-modality approaches. In particular, the accuracy of single-modality MRI model was 93.95% and adding cognitive information with an attention-based fusion technique improved accuracy up to 98.85%. Also, the system showed good results in differentiating between Normal (CN) and EMCI that is considered crucial stage for medical intervention but hard to distinguish visually.

### **6.2 Scope and Objective Assessment**

The project successfully fulfilled the defined scope, meeting all core objectives through a combination of computational research and software development:

1. **Multimodal System Creation:** We proposed and validated a deep learning framework for the effective integration of structural MRI data with cognitive assessment markers (MoCA, ADAS-13, and CDR-SB).
2. **Comparative Analysis:** A detailed evaluation of different architectures (CNN-only vs Multimodal Fusion) was done using accuracy, precision, recall and F1-score as performance metrics to ensure the accuracy of the analysis.

3. **User Interface Development:** We have created an effective interface that bridges the gap between algorithmic research and usability. It lets the user to input MRI scans and cognitive data to automatically predict diagnosis.
4. **Diagnostic Classification:** The system was able to classify subjects into CN, EMCI, LMCI and AD, and can be used as a reliable diagnostic support tool in an academic research setting .

## **6.3 Challenges Faced**

The development process involved several technical and logical hurdles:

1. **Data Alignment:** The challenge was to do the "Patient-Wise Alignment" correct, as each MRI scan had to be aligned correctly with the corresponding cognitive scores in the ADNI database.
2. **Holistic Integration:** We had to carefully modify the Attention Fusion head to balance the “weight” of the MRI features to the cognitive vector to ensure that one data mode didn’t take over the other.
3. **Architecture Complexity:** Building an InceptionV3 and MobileNet ensemble required a lot of optimization to run efficiently within the available computational resources.
4. **Preprocessing 3D Data:** A major technical challenge was the conversion of 3D NIfTI neuroimaging data to standardized 2D axial slices that kept the structural integrity of the hippocampus and other regions of focus.

## **6.4 Limitations**

1. The limitation is that a user interface is developed but the system has not been medically tested in real time in a hospital or live trial with patients.
2. The project was limited to diagnostic classification. It did not relate to disease prognosis, or model the longitudinal progression of disease over time.

## **6.5 Future Work**

To build upon the foundation laid by this research, the following directions are recommended:

1. **Clinical Validation:** Moving the system into a healthcare facility to verify it in real-time with medical professionals and evaluate its practicality as a diagnostic assistant.
2. **Longitudinal Modeling:** Expanding the scope by including “Prognosis” where the model can predict when a patient may switch from EMCI to AD.
3. **Explainable AI (XAI) Enhancements:** Improve the user interface to show detailed heatmaps of brain atrophy in order to help clinicians understand the “Why” behind each AI forecast.

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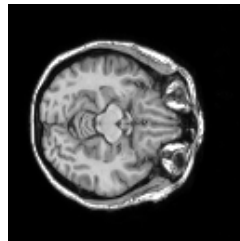
## **Appendix**

### **Appendix A: Data Set Examples**

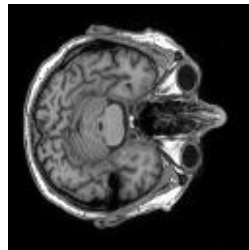
The data set used in this research study is made up of MRI data and Cognitive Dataset. Both are from Alzheimer's Disease Neuroimaging Initiative (ADNI). The MRI data set has been classified into four types:

1. Cognitively Normal (CN)
2. Early Mild Cognitive Impairment (EMCI)
3. Late Mild Cognitive Impairment (LMCI)
4. Alzheimer's Disease (AD)

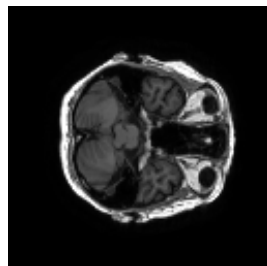
All the four classes have MRI images. Below are examples of each class:



**Appendix A.1 EMCI Sample**



**Appendix A.2 LMCI Sample**



**Appendix A.3 AD Sample**

The Cognitive dataset consisted of following sections:

1. RID
2. VISCODE
3. DX
4. MOCA
5. CDRSB
6. ADAS13

## Appendix B: Multimodal Model Code Snippet

```
# =====
# LIBRARIES
# =====
import os
import numpy as np
import pandas as pd
import tensorflow as tf
from tensorflow.keras import layers, Model
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from tensorflow.keras.optimizers import Adam
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.utils.class_weight import compute_class_weight
import matplotlib.pyplot as plt
import seaborn as sns

from google.colab import drive
drive.mount('/content/drive')

# =====
# PATHS
# =====
BASE_DIR = "/content/drive/MyDrive/FYP_Dataset"
COG_PATH = "/content/drive/MyDrive/FYP_Dataset/cognitive_dataset.csv"
SAVE_PATH =
"/content/drive/MyDrive/FYP_Dataset/incv3_mobnet_cog30_fusion_model.h5"

# =====
# LOAD MRI FEATURES
```

```

# =====
def load_split(split):
    X = np.load(os.path.join(BASE_DIR, f"incv3_mobnet_features_{split}.npy"))
    ids = np.load(os.path.join(BASE_DIR,
f"incv3_mobnet_patient_ids_{split}.npy"))
    return X, ids

X_mri_train, ids_train = load_split("train")
X_mri_val, ids_val = load_split("val")
X_mri_test, ids_test = load_split("test")

print("✓ MRI features loaded")

# =====
# EXTRACT NUMERIC PATIENT ID
# =====
def extract_patient_number(mri_id):
    try:
        return int(mri_id.split('_S_')[1].split('_')[0])
    except:
        return None

ids_train_num = np.array([extract_patient_number(pid) for pid in ids_train])
ids_val_num = np.array([extract_patient_number(pid) for pid in ids_val])
ids_test_num = np.array([extract_patient_number(pid) for pid in ids_test])

# =====
# LOAD & PREPROCESS COGNITIVE DATA
# =====
cog_df = pd.read_csv(COG_PATH)

le = LabelEncoder()
cog_df["label_enc"] = le.fit_transform(cog_df["DX"])

numeric_features = ["MOCA", "CDRSB", "ADAS13"]
cog_df = cog_df.dropna(subset=numeric_features)

scaler = StandardScaler()
cog_df[numeric_features] = scaler.fit_transform(cog_df[numeric_features])

```

```

cog_df["RID"] = cog_df["RID"].astype(int)

print("✓ Cognitive data loaded & normalized")

# =====
# GENERATE 30-QUESTION VECTOR
# =====
NUM_QUESTIONS = 30

def generate_question_vector(row):
    q = np.zeros(NUM_QUESTIONS, dtype=np.float32)

    # Memory (Q1-Q8)
    if row["MOCA"] < -0.5:
        q[0:8] = 1

    # Attention (Q9-Q14)
    if row["MOCA"] < 0:
        q[8:14] = 1

    # Executive Function (Q15-Q20)
    if row["ADAS13"] > 0.5:
        q[14:20] = 1

    # Language (Q21-Q24)
    if row["ADAS13"] > 0:
        q[20:24] = 1

    # Orientation (Q25-Q27)
    if row["MOCA"] < -1.0:
        q[24:27] = 1

    # Daily Functioning (Q28-Q30)
    if row["CDRSB"] > 0:
        q[27:30] = 1

    return q

```



```

# =====
# ALIGN MRI + COGNITIVE (PATIENT-WISE)
# =====
def align_data(mri_X, mri_ids, split_name):
    X_mri, X_cog, y = [], [], []
    matched = 0

    for i, pid in enumerate(mri_ids):
        rows = cog_df[cog_df["RID"] == pid]
        if len(rows) == 0:
            continue

        row = rows.iloc[0]

        matched += 1
        X_mri.append(mri_X[i])
        X_cog.append(generate_question_vector(row))
        y.append(row["label_enc"])

    print(f"✓ {split_name}: Matched {matched}/{len(mri_ids)} patients")
    return (
        np.array(X_mri, dtype=np.float32),
        np.array(X_cog, dtype=np.float32),
        np.array(y, dtype=np.int32)
    )

X_mri_train, X_cog_train, y_train = align_data(X_mri_train, ids_train_num,
"TRAIN")
X_mri_val, X_cog_val, y_val = align_data(X_mri_val, ids_val_num, "VAL")
X_mri_test, X_cog_test, y_test = align_data(X_mri_test, ids_test_num, "TEST")

# =====
# CLASS WEIGHTS
# =====
class_weights = compute_class_weight(
    class_weight="balanced",
    classes=np.unique(y_train),
    y=y_train
)

```

```

class_weights = dict(enumerate(class_weights))
print("Class weights:", class_weights)

# =====
# MULTIMODAL ATTENTION FUSION MODEL
# =====
# MRI branch
mri_input = layers.Input(shape=(X_mri_train.shape[1],), name="MRI_Input")
mri_feat = layers.Dense(256, activation="relu")(mri_input)
mri_feat = layers.BatchNormalization()(mri_feat)
mri_feat = layers.Dropout(0.4)(mri_feat)

# Cognitive branch (30 questions)
cog_input = layers.Input(shape=(30,), name="Cognitive_Input")
cog_feat = layers.Dense(128, activation="relu")(cog_input)
cog_feat = layers.BatchNormalization()(cog_feat)
cog_feat = layers.Dropout(0.3)(cog_feat)
cog_feat = layers.Dense(256, activation="relu")(cog_feat)

# Attention Fusion
stacked = layers.Concatenate(axis=1)([
    layers.Reshape((1, 256))(mri_feat),
    layers.Reshape((1, 256))(cog_feat)
])

attention = layers.Dense(1, activation="softmax")(stacked)
attended = layers.Multiply()([stacked, attention])
attended = layers.Flatten()(attended)

fusion = layers.Dense(128, activation="relu")(attended)
fusion = layers.BatchNormalization()(fusion)
fusion = layers.Dropout(0.5)(fusion)

output = layers.Dense(len(le.classes_), activation="softmax")(fusion)

fusion_model = Model([mri_input, cog_input], output)

fusion_model.compile(
    optimizer=Adam(1e-4),
    loss="sparse_categorical_crossentropy",

```

```

        metrics=["accuracy"]
    )

fusion_model.summary()

# =====
# TRAIN
# =====
callbacks = [
    EarlyStopping(patience=10, restore_best_weights=True),
    ReduceLROnPlateau(patience=5, factor=0.3, min_lr=1e-6)
]

history = fusion_model.fit(
    [X_mri_train, X_cog_train],
    y_train,
    validation_data=([X_mri_val, X_cog_val], y_val),
    epochs=60,
    batch_size=32,
    class_weight=class_weights,
    callbacks=callbacks
)

# =====
# SAVE MODEL
# =====
fusion_model.save(SAVE_PATH, include_optimizer=False)
print(f"✓ Model saved at {SAVE_PATH}")

# =====
# EVALUATION
# =====
y_pred = np.argmax(fusion_model.predict([X_mri_test, X_cog_test]), axis=1)

print("Test Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))

# =====
# TRAINING CURVES

```

```

# =====
plt.figure(figsize=(8,5))
plt.plot(history.history["accuracy"], label="Train Accuracy")
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.title("Training vs Validation Accuracy")
plt.legend()
plt.grid(True)
plt.show()

plt.figure(figsize=(8,5))
plt.plot(history.history["loss"], label="Train Loss")
plt.plot(history.history["val_loss"], label="Validation Loss")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend()
plt.grid(True)
plt.show()

# =====
# CONFUSION MATRIX (HEATMAP)
# =====
y_pred = np.argmax(
    fusion_model.predict([X_mri_test, X_cog_test]),
    axis=1
)

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(7,6))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=le.classes_,
    yticklabels=le.classes_
)

```

```
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix - Multimodal Fusion Model")
plt.tight_layout()
plt.show()

print("Test Accuracy:", accuracy_score(y_test, y_pred))
```